

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 4 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|-----------------------------------|--|
| 1. Rational function _____ | (A) A value that makes the <i>original</i> denominator zero, so it can never be an input. |
| 2. Domain restriction _____ | (B) A horizontal line the graph approaches at its far left and far right; found by comparing the degrees of the numerator and denominator. |
| 3. Vertical asymptote _____ | (C) The number k in $y = kx$ or $y = \frac{k}{x}$; you find it from one known pair of values. |
| 4. Hole _____ | (D) A quotient of two polynomials whose denominator is not zero. |
| 5. Horizontal asymptote _____ | (E) A single missing point on an otherwise unbroken curve, left behind by a factor that cancels. |
| 6. Least common denominator _____ | (F) A number that turns up when you solve but must be thrown out, because it is an excluded value of the original equation. |
| 7. Extraneous solution _____ | (G) The smallest expression that every denominator divides evenly, built from the factored denominators. |
| 8. Constant of variation _____ | (H) A vertical line the graph runs alongside but never touches, left by a factor that does <i>not</i> cancel. |

Part B — Multiple Choice (12 pts)

Circle the best answer.

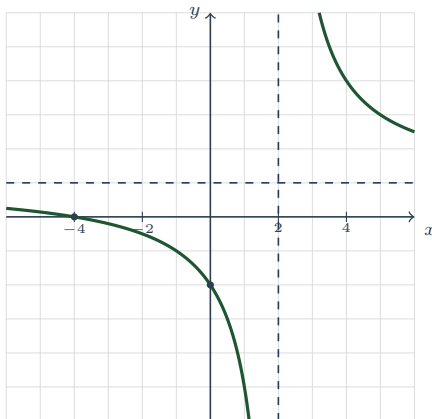
- The domain of $f(x) = \frac{x+5}{x^2-9}$ is all real numbers except (A) $x = -5$ (B) $x = 3$ (C) $x = \pm 3$ (D) none — all reals
- Simplified completely, $\frac{x^2-25}{x^2+3x-10}$ is (A) $\frac{x-5}{x-2}, x \neq 2$ (B) $\frac{x-5}{x-2}, x \neq -5, 2$ (C) $\frac{x+5}{x-2}, x \neq 2$ (D) $\frac{-5}{3x-10}$
- The horizontal asymptote of $f(x) = \frac{3x^2-1}{x^2+4x}$ is (A) $y = 0$ (B) $y = \frac{1}{3}$ (C) $y = 3$ (D) there is none
- Which statement about $h(x) = \frac{x^2-x-6}{x^2-9}$ is true?
 - There are vertical asymptotes at $x = 3$ and at $x = -3$.
 - There is a hole at $x = 3$ and a vertical asymptote at $x = -3$.
 - There is a hole at $x = -3$ and a vertical asymptote at $x = 3$.
 - There is a hole at $x = -2$ and a vertical asymptote at $x = 3$.

5. The graph of $g(x) = \frac{1}{x-4} + 2$ has asymptotes (A) $x = 4, y = 2$ (B) $x = -4, y = 2$ (C) $x = 4, y = 0$ (D) $x = 2, y = 4$
6. y varies *inversely* as x , and $y = 8$ when $x = 3$. When $x = 12$, $y =$ (A) 2 (B) 6 (C) 24 (D) 32

Part C — Short Answer & Computation (40 pts)

Show your work. State every restriction on the variable.

1. **Read the graph.** The rational function f is graphed below; the dashed lines are its asymptotes and each gridline is one unit.



- (a) Equation of the vertical asymptote: _____ Equation of the horizontal asymptote: _____
- (b) x -intercept: _____; y -intercept: _____
- (c) Domain (interval notation): _____
- (d) Range (interval notation): _____
- (e) End behavior: as $x \rightarrow \pm\infty$, $f(x) \rightarrow$ _____
- (f) Interval(s) where f is decreasing: _____
- (g) Interval(s) where $f(x) > 0$: _____
2. **Simplify and state the restrictions.** Read the restrictions from the *original* denominator.
- (a) $\frac{3x^2 - 27}{x^2 + x - 12}$ (b) $\frac{x^2 - 16}{4 - x}$

3. **Multiply or divide.** Factor first; give the simplest form with its restrictions.

(a) $\frac{x^2 - 4}{x^2 + 6x + 9} \cdot \frac{x + 3}{x - 2}$

(b) $\frac{x^2 - x - 6}{x^2 - 1} \div \frac{x - 3}{x + 1}$

4. **Add, subtract, and simplify a complex fraction.**

(a) $\frac{5}{x + 2} - \frac{3}{x - 1}$

(b) $\frac{x}{x^2 - 9} + \frac{2}{x + 3}$

(c) $\frac{\frac{1}{x} + \frac{1}{2}}{\frac{1}{x} - \frac{1}{2}}$

5. **The parent function and its transformations.** Let $g(x) = \frac{1}{x + 3} - 2$.

(a) Describe the transformation(s) that take $y = \frac{1}{x}$ to g . _____

(b) Vertical asymptote: _____; horizontal asymptote: _____

(c) Domain: _____; range: _____

(d) A different hyperbola of the form $y = \frac{1}{x - h} + k$ has vertical asymptote $x = 1$ and horizontal asymptote $y = 4$. Write its equation. _____

6. **Analyze from the equation.** For $f(x) = \frac{2x^2 - 8}{x^2 - x - 6}$:

(a) Factor the numerator and denominator, and list every excluded value.

(b) Which excluded value gives a *hole* and which gives a *vertical asymptote*? Give the coordinates of the hole.

(c) Horizontal asymptote: _____ (Justify with a degree comparison.)

(d) x -intercept: _____; y -intercept: _____

7. **Solve. Check every answer against the excluded values.**

(a) $\frac{x}{x+2} + \frac{1}{x-1} = 1$

(b) $\frac{x}{x-4} = \frac{4}{x-4} + 3$

(c) $\frac{x}{x-2} + \frac{2}{x} = \frac{4}{x^2 - 2x}$

8. **Variation.** The table below records a pair of related quantities.

x	2	4	5	10
y	30	15	12	6

(a) Is y directly or inversely proportional to x ? Give the computation from the table that proves it.

(b) Find the constant of variation k and write the equation. _____

(c) Use your equation to find y when $x = 8$. _____

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **The full analysis.** Consider $f(x) = \frac{x^2 - 5x + 4}{x^2 - 16}$. **(a)** Factor both parts and list every excluded value of the *original* function. **(b)** Run the cancel test: which excluded value is a hole and which is a vertical asymptote? Give the hole's coordinates and explain how you found its height. **(c)** State the horizontal asymptote, justify it with a degree comparison, and write the end behavior it describes. **(d)** Give the x - and y -intercepts. **(e)** Write the domain in interval notation. **(f)** A classmate simplifies f to $\frac{x-1}{x+4}$ and says the two functions are the same. Say exactly where they agree and where they do not.

2. **Modeling.** A club pays a one-time setup fee of \$180 to a screen printer, plus \$7 for each shirt printed. The *average cost per shirt* for an order of n shirts is

$$A(n) = \frac{7n + 180}{n}.$$

- (a) Explain why A is a rational function, and state the domain that makes sense for this situation. (b) Find $A(20)$ and $A(60)$, with units. What is happening to the average cost? (c) Rewrite $A(n)$ in the form $7 + \frac{180}{n}$ and use it to state the horizontal asymptote. What does that number mean about this order? (d) Find the order size for which the average cost is exactly \$9 per shirt. (e) Can the average cost ever be exactly \$7? Answer using the horizontal asymptote.